Tech Medium

Dimension	Question	Answer
AI Heroes Universe	What model uses diffusion noise-denoise loops to generate images?	Generative AI
AI Heroes Universe	What architecture uses competing generator and discriminator networks?	Generative AI
AI Heroes Universe	What framework combines ReAct prompting loops with tool execution?	Agentic AI
AI Heroes Universe	What autonomous AI breaks goals into sub-tasks using feedback?	Agentic AI
AI Heroes Universe	What algorithm combines L1 and L2 penalties to prevent overfitting?	Machine Learning
AI Heroes Universe	What ensemble method trains sequential trees that correct prior errors?	Machine Learning
AI Heroes Universe	What method computes gradients via chain rule during backpropagation?	Deep Learning
AI Heroes Universe	What neural architecture uses self-attention to process sequential data parallelly?	Deep Learning
AI Heroes Universe	What embeddings capture contextual word meanings, like BERT or Word2Vec?	NLP
AI Heroes Universe	What metric measures n-gram overlap to evaluate translation quality?	NLP
AI Heroes Universe	What technique breaks complex reasoning into step-by-step logical thoughts?	Prompt Engineering
AI Heroes Universe	What method optimizes prompts dynamically using automated feedback loops?	Prompt Engineering

AI Heroes Universe	What indexing algorithm uses small-world graphs for fast similarity search?	Vector Database
AI Heroes Universe	What metric measures the angle between vectors for semantic similarity?	Vector Database
AI Heroes Universe	What hybrid search combines BM25 keyword matching with vector similarity?	RAG
AI Heroes Universe	What optimization re-ranks retrieved document chunks before LLM context injection?	RAG
AI Heroes Universe	What PEFT method freezes weights and injects low-rank trainable matrices?	LLM
AI Heroes Universe	What alignment method optimizes language models using human preference pairs?	LLM
AI Heroes Universe	What pattern gives software agents long-term memory via vector stores?	AI Agent
AI Heroes Universe	What multi-agent framework enables conversational personas to collaborate on code?	AI Agent
AI Heroes Universe	What matrix factorization uncovers latent factors for personalized product ranking?	Recommendation Engine
AI Heroes Universe	What cold-start strategy combines user demographic metadata with item attributes?	Recommendation Engine
AI Heroes Universe	What single-stage object detector processes images in one forward pass?	Computer Vision
AI Heroes Universe	What segmentation network uses skip connections to preserve spatial details?	Computer Vision
AI Heroes Universe	What time-series model combines autoregressive, differencing, and moving average components?	Predictive Analytics
AI Heroes Universe	What statistical model estimates hazard rates while handling right-censored data?	Predictive Analytics

AI Heroes Universe	What framework uses Shapley values from game theory for feature attribution?	Responsible AI
AI Heroes Universe	What auditing evaluates models across demographic subgroups to prevent bias?	Responsible AI
AI Heroes Universe	What dialog component tracks conversation state and slots across turns?	Chatbot
AI Heroes Universe	What fallback mechanism handles low-confidence intent classification in chatbots?	Chatbot
Programming Powers	What feature wraps functions to dynamically extend behavior using `@`?	Python
Programming Powers	What memory-efficient construct yields values lazily using `yield`?	Python
Programming Powers	What module provides event loops and coroutines using `async`/`await`?	Python
Programming Powers	What analytical functions perform calculations across related rows using `OVER()`?	SQL
Programming Powers	What database index structure uses B-trees to accelerate `SELECT` queries?	SQL
Programming Powers	What transaction property guarantees all operations complete or roll back?	SQL
Programming Powers	What mechanism automatically reclaims unreferenced heap memory objects at runtime?	Java
Programming Powers	What design feature allows classes to implement multiple interface contracts?	Java
Programming Powers	What framework component provides dependency injection to wire enterprise beans?	Java
Programming Powers	What protocol enforces XML schemas, SOAP headers, and ACID compliance?	API
Programming Powers	What rate-limiting algorithm uses fixed capacity tokens to control traffic spikes?	API

Programming Powers	What Python library handles data validation and type enforcement in FastAPI?	FastAPI
Programming Powers	What ASGI server provides high-performance asynchronous networking for FastAPI apps?	FastAPI
Programming Powers	What schema standard validates structural formats and data types of JSON?	JSON
Programming Powers	What token format encodes signed claims as base64 JSON objects?	JSON
Programming Powers	What command applies a specific commit from another branch directly?	Git
Programming Powers	What command re-applies commits on top of another base tip?	Git
Programming Powers	What command temporarily shelves uncommitted local working directory changes?	Git
Programming Powers	What automation feature runs custom CI/CD YAML workflows on repo push?	GitHub
Programming Powers	What security feature scans repository commit histories for exposed API tokens?	GitHub
Programming Powers	What special syntax starting with `%` executes cell environment configurations?	Jupyter Notebook
Programming Powers	What kernel process executes code cell commands in Jupyter notebooks?	Jupyter Notebook
Programming Powers	What widget library creates sliders and UI controls inside Jupyter cells?	Jupyter Notebook
Programming Powers	What constraint specifies that REST APIs must not store client session state?	REST API
Programming Powers	What HTTP status code indicates successful resource creation during POST?	REST API
Programming Powers	What maturity model categorizes REST APIs using URIs, verbs, and HATEOAS?	REST API

Programming Powers	What practice replaces external dependencies with controlled mocks during unit tests?	Unit Testing
Programming Powers	What metric measures the percentage of decision branches executed during tests?	Unit Testing
Programming Powers	What interactive tool pauses execution to inspect stack frames and variables?	Debugging
Programming Powers	What technique detects memory leaks by analyzing object allocations in heaps?	Debugging
Cloud Universe	What serverless compute service automatically runs code without managing EC2 instances?	AWS
Cloud Universe	What cloud network isolates virtual networks using private subnets and NAT gateways?	AWS
Cloud Universe	What database service offers multi-AZ replication for MySQL and PostgreSQL?	AWS
Cloud Universe	What identity service handles authentication and single sign-on across Azure resources?	Azure
Cloud Universe	What serverless integration service builds event-driven workflows connecting cloud apps?	Azure
Cloud Universe	What multi-model database offers guaranteed single-digit millisecond latency globally?	Azure
Cloud Universe	What serverless data warehouse runs SQL queries across petabytes using columnar storage?	GCP
Cloud Universe	What managed container platform runs stateless apps, auto-scaling down to zero?	GCP
Cloud Universe	What unified ML platform provides tools for dataset management and model deployment?	GCP
Cloud Universe	What build technique uses multiple `FROM` statements to reduce image size?	Docker

Cloud Universe	What feature persists container data by mounting host directories directly?	Docker
Cloud Universe	What multi-container tool uses YAML files to run application stacks?	Docker
Cloud Universe	What API object routes external HTTP/HTTPS traffic to cluster services?	Kubernetes
Cloud Universe	What controller automatically scales pod replicas based on CPU utilization?	Kubernetes
Cloud Universe	What abstraction provisions persistent storage dynamically via PersistentVolumeClaims in Kubernetes?	Kubernetes
Cloud Universe	What release strategy deploys software updates to a small user subset first?	DevOps
Cloud Universe	What deployment pattern uses two identical environments for zero-downtime releases?	DevOps
Cloud Universe	What monitoring practice collects metrics, logs, and traces for system visibility?	DevOps
Cloud Universe	What artifact repository securely stores and scans Docker images for vulnerabilities?	CI/CD
Cloud Universe	What pipeline feature executes automated test suites across parallel worker nodes?	CI/CD
Cloud Universe	What GitOps approach uses Git repos as single source of truth for infrastructure?	CI/CD
Cloud Universe	What Linux kernel feature isolates process trees and mount points for containers?	Containerization
Cloud Universe	What Linux kernel mechanism limits CPU and memory usage for container groups?	Containerization
Cloud Universe	What standard specification defines container runtime execution and image formats?	Containerization

Cloud Universe	What central file maps real-world cloud resources to declarative configuration code?	Terraform
Cloud Universe	What reusable configuration structure packages standardized cloud provisioning code patterns?	Terraform
Cloud Universe	What preview command compares declarative code against remote cloud infrastructure?	Terraform
Cloud Universe	What model provisions elastic compute resources dynamically matching traffic shifts?	Cloud Computing
Cloud Universe	What model divides security responsibilities between cloud provider and client?	Cloud Computing
Cloud Universe	What deployment strategy distributes workloads across multiple isolated availability zones?	Cloud Computing
Analytics Dimension	What reporting metric measures the percentage of user interactions achieving conversion?	Dashboard
Analytics Dimension	What UI feature filters multiple dashboard widgets simultaneously via selected parameters?	Dashboard
Analytics Dimension	What capability allows navigating from summary metrics down to granular transactions?	Dashboard
Analytics Dimension	What functional query language constructs calculated columns and measures in Power BI?	Power BI
Analytics Dimension	What engine inside Power BI cleans and shapes data using M language?	Power BI
Analytics Dimension	What feature automatically refreshes published report datasets on scheduled intervals?	Power BI
Analytics Dimension	What feature computes aggregates at specified dimensions independent of view granularity?	Tableau

Analytics Dimension	What file format packages data extracts and formatting for offline sharing?	Tableau
Analytics Dimension	What data structure combines tables flexibly without requiring physical JOIN operations?	Tableau
Analytics Dimension	What narrative structure arranges analytical insights into context, conflict, and resolution?	Data Storytelling
Analytics Dimension	What design principle eliminates chartjunk to maximize data-ink display ratio?	Data Storytelling
Analytics Dimension	What cognitive principle arranges visual elements to match human perceptual grouping?	Data Storytelling
Analytics Dimension	What framework aligns strategic business goals with key results over time?	KPI
Analytics Dimension	What leading metric measures early operational activity to predict future outcomes?	KPI
Analytics Dimension	What metric evaluates overall customer satisfaction based on recommendation likelihood?	KPI
Analytics Dimension	What unsupervised clustering algorithm partitions customer data into k distinct clusters?	Customer Segmentation
Analytics Dimension	What framework categorizes customer value using Recency, Frequency, and Monetary metrics?	Customer Segmentation
Analytics Dimension	What survival analysis model estimates hazard rates to identify cancellation risks?	Churn Analysis
Analytics Dimension	What predictive approach calculates customer retention probabilities to trigger automated re-engagement?	Churn Analysis
Analytics Dimension	What cohort visualization tracks user group retention rates over consecutive intervals?	Churn Analysis

Analytics Dimension	What p-value threshold indicates probability that conversion differences occurred by chance?	A/B Testing
Analytics Dimension	What sample size calculation determines minimum traffic required for statistical significance?	A/B Testing
Analytics Dimension	What statistical error occurs when an experiment falsely claims a variant won?	A/B Testing
Analytics Dimension	What formula combines average order value, purchase frequency, and churn rate?	CLV
Analytics Dimension	What discount calculation incorporates time value of money when projecting future cash flows?	CLV
Analytics Dimension	What capital metric calculates the discount rate making Net Present Value zero?	ROI
Analytics Dimension	What financial metric calculates the duration needed to recover initial investment costs?	ROI
Analytics Dimension	What multivariate chart displays quantitative data across x-axis, y-axis, and bubble size?	Data Visualization
Analytics Dimension	What plot displays median, quartiles, and statistical outliers across numerical distributions?	Data Visualization
Analytics Dimension	What graphical technique uses color intensity gradients to display matrix values?	Data Visualization
Emerging Technology Zone	What consensus algorithm requires network nodes to stake crypto assets as collateral?	Blockchain
Emerging Technology Zone	What self-executing contract stored on a blockchain enforces agreement terms automatically?	Blockchain
Emerging Technology Zone	What data structure uses binary trees of hashes to verify transaction integrity?	Blockchain

Emerging Technology Zone	What lightweight publish-subscribe protocol is optimized for low-bandwidth smart devices?	IoT
Emerging Technology Zone	What architecture processes telemetry locally on smart gateways near sensors?	IoT
Emerging Technology Zone	What virtual digital replica mirrors physical hardware assets in real time?	IoT
Emerging Technology Zone	What monitoring process detects input data distribution shifts that degrade model accuracy?	MLOps
Emerging Technology Zone	What central repository manages and serves curated features for real-time inference?	MLOps
Emerging Technology Zone	What deployment pattern routes traffic to multiple model versions to compare performance?	MLOps
Emerging Technology Zone	What feature connects visual low-code interfaces to custom REST web services?	Low Code
Emerging Technology Zone	What governance framework manages security permissions and lifecycles for low-code apps?	Low Code
Emerging Technology Zone	What visual engine models business logic step-by-step using flowchart decision nodes?	Low Code
Emerging Technology Zone	What visual database modeling approach defines table schemas without SQL code?	No Code
Emerging Technology Zone	What integration platform connects separate SaaS apps via webhooks without programming?	No Code
Emerging Technology Zone	What responsive UI container adjusts component layout across mobile and desktop displays?	No Code
Emerging Technology Zone	What technology combines OCR with deep learning to extract data from invoices?	Intelligent Automation

Emerging Technology Zone	What bot automation approach executes front-end UI interactions without changing backend systems?	Intelligent Automation
Emerging Technology Zone	What process mining technique analyzes event logs to map workflow bottlenecks?	Intelligent Automation
Emerging Technology Zone	What contextual window allows coding assistants to examine open project files for completions?	Copilot
Emerging Technology Zone	What inline developer feature allows programmers to request automated refactoring via chat?	Copilot
Emerging Technology Zone	What security filter prevents AI assistants from outputting protected open-source snippets?	Copilot
Emerging Technology Zone	What search architecture combines sparse lexical scores with dense vector semantic scores?	Semantic Search
Emerging Technology Zone	What NLU model generates dense vector embeddings that encode whole sentence intent?	Semantic Search
Emerging Technology Zone	What re-ranking mechanism evaluates search results using deep cross-attention models?	Semantic Search
Emerging Technology Zone	What joint embedding architecture maps images and text into a shared vector space?	Multimodal AI
Emerging Technology Zone	What vision-language architecture accepts image and text inputs to answer visual questions?	Multimodal AI
Emerging Technology Zone	What cross-modal fusion strategy combines visual feature maps and audio spectrum embeddings?	Multimodal AI
Emerging Technology Zone	What quantum phenomenon binds two qubits so one state dictates the other?	Quantum Computing

Emerging Technology Zone	What quantum principle allows qubits to exist in 0 and 1 states simultaneously?	Quantum Computing
Emerging Technology Zone	What quantum algorithm provides quadratic speedup for searching unstructured databases?	Quantum Computing